RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
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Department of Artificial Intelligence and Machine Learning

Date: 14.01.2026	Improvement Test	Max. Marks: 50+10
Semester: V	UG	Duration: $1\frac{1}{2}$ Hrs + ½ Hr
Title: Mathematical Algorithms for Artificial Intelligence		Course Code: AI255BTC

Part A

Q. No		PART A Quiz 1	M	BT	CO
1	a)	Which among the following can be the characteristic equation for a REAL SYMMETRIC MATRIX? a) $\lambda(\lambda^2+4)(\lambda+3)=0$ b) $\lambda(\lambda^4+16)=0$ c) $\lambda(\lambda^2-9)=0$ d) None of the above can be the characteristic equation for a real symmetric matrix	2	1	1
	b)	Let A be a 3x3 real matrix of rank 1. You are told that the trace of the matrix is 9. The eigenvalues of A are a) 0, 0, 3 b) 0, 0, -3 c) 0, 0, 9 d) 0, 0, -9	2	1	2
	c)	Given the matrix $A=\begin{pmatrix}1\\1\\2\end{pmatrix}$, the pseudoinverse of this matrix is a) Non-existent for A, as it is a single column matrix b) $(A^TA)^{-1}A^T$ c) $A^T(AA^T)^{-1}$ d) A^{-1} $-\lambda^3-9\lambda^2+9\lambda+0$ $\lambda(\lambda$ $\lambda(\lambda^2+9)=0$	2	2	3
	d)	The eigenvalues of a projection matrix can only be a) ± 1 b) 0 and 1 c) 0 and -1 d) Determined based on the matrix.	2	2	2
	e)	Statement 1: Gram-Schmidt Orthonormalization always yields orthonormal vectors Statement 2: Orthonormal vectors are always linearly independent. a) Statement 1 and Statement 2 are ALWAYS TRUE b) Statement 1 and Statement 2 are ALWAYS FALSE c) Only Statement 1 is ALWAYS TRUE d) Only Statement 2 is ALWAYS TRUE	2	2	3

P_1

$a-\lambda$ c
c $b-\lambda$ $ab-c^2$
$(a-\lambda)(b-\lambda)-c^2$

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SL. No	Part B Questions	M	BT	CO
1.	Suppose A is an $n \times n$ orthogonal matrix, and $x \in \mathbb{R}^n$ prove or disprove that $\|\|Ax\|\| = \|\|x\|\|$, where $\|\| \ \|\|$ is the length of the vector. *No partial marks will be given if you take a specific numerical problem and solve.*	8	4	2
2.	Let u and v be two vectors in $\mathbb{R}^2$, such that $A = uv^T$. Obtain the SVD of the matrix A. *No partial marks will be given if you take a specific numerical problem and solve.*	8	3	2
3.	Given a singular value decomposition of an $m \times n$ matrix $A = U\Sigma V^T$, find an SVD of A^T. How are the singular values of A^T related to A?	8	3	2
4.	Show that if x is in both W and $W^{\perp}$, $x = 0$. *No partial marks will be given if you take a specific numerical problem and solve.*	8	4	2
5.	If P is a projection matrix, prove or disprove (I-P) is also a projection matrix, where I is the identity matrix. *Hint: Recall the definition of Projection matrix.*	8	4	2
6.	Let $M = I - 2uu^T$ where u is a unit vector in $\mathbb{R}^2$. This matrix M reflects every vector about the line passing through the origin and along the direction of u. Obtain the eigenvalues of the matrix M. OR (a) Prove that for a real symmetric matrix A, the eigenvectors corresponding to distinct eigenvalues are orthogonal to each other. (b) Obtain the conditions for solution to exist for the system of linear equations $Ax = b$. Assume A is a tall matrix with rank $r < n$, where n is the number of columns of A.	10	4	2